

Chi-Squared Test of Independence

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Introduction

The chi-squared test of independence asks whether two categorical variables in a contingency table are associated. Under the null of independence, cell counts follow a predictable pattern determined by the row and column margins; large departures constitute evidence of association.

Prerequisites

Contingency tables, goodness-of-fit chi-squared.

Theory

For an $r \times c$ table with observed counts O_{ij} , row totals R_i , column totals C_j , and grand total n , the expected count under independence is

$$E_{ij} = R_i C_j / n.$$

Test statistic:

$$\chi^2 = \sum_{i,j} \frac{(O_{ij} - E_{ij})^2}{E_{ij}} \sim \chi^2_{(r-1)(c-1)} \quad \text{under } H_0.$$

Effect sizes:

- 2x2: Phi $\phi = \sqrt{\chi^2/n}$, equivalent to the Matthews correlation.
- $r \times c$: Cramer's $V = \sqrt{\chi^2/[n \min(r-1, c-1)]}$; range $[0, 1]$.
- Thresholds: 0.10 / 0.30 / 0.50 for small / medium / large.

Assumptions

- Independent observations.
- Expected counts ≥ 5 in every cell (else Fisher's exact or Monte Carlo chi-squared).

R Implementation

```
library(vcd); library(effectsize)

tab <- matrix(c(42, 38, 65,
                28, 55, 50,
                18, 32, 40), nrow = 3, byrow = TRUE,
              dimnames = list(Stage = c("I", "II", "III"),
                              Site  = c("A", "B", "C")))

tab
```

```
chi <- chisq.test(tab)
chi
chi$expected
chi$stdres      # standardised residuals

# Effect sizes
cramers_v(tab)
assocstats(tab)

# Mosaic plot of residuals
mosaic(tab, shade = TRUE, legend = TRUE)
```

Output & Results

Pearson's Chi-squared test
X-squared = 8.92, df = 4, p-value = 0.063

Cramer's V | 95% CI

0.128 | [0.00, 0.21]

Borderline rejection at $\alpha = 0.05$. Small effect by Cramer's V. Standardised residuals identify which cells drive the pattern.

Interpretation

“Stage and site were not significantly associated ($\chi^2(4) = 8.92$, $p = 0.063$, Cramer's V = 0.13).” Report counts and row/column percentages alongside the test.

Practical Tips

- For 2x2 tables with any expected count below 5, use Fisher's exact test.
- For larger tables with sparse cells, use `simulate.p.value = TRUE` for a Monte Carlo p-value.
- Report effect size (Cramer's V) alongside the test; significance alone is uninformative for large n .
- Yates' continuity correction is available but rarely necessary; omit it via `correct = FALSE`.
- For ordered categorical variables, use the Cochran-Armitage trend test or the linear-by-linear association test.

For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren't.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

See also — labs in this chapter

- Populations, samples, and the central limit theorem
- Bootstrap and permutation tests
- Maximum likelihood estimation
- One-sample t-test and one-proportion test
- Hypothesis testing, p-values, type I/II errors

- Two-sample and paired t-tests
- Two proportions, chi-square, risk and odds
- Pearson and Spearman correlation
- Non-parametric tests